

The equivalent circuit method

Testing the *ECM_MS* software using 3D magnetostatics examples

https://github.com/JNSresearcher/ECM_MS

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Ferromagnetic regions with saturation

Testing was conducted for research purposes, to evaluate the algorithm's performance, debug the code, etc.

Test 1

For validation, a task similar to the reference task 20 for the TEAM workshop was [1]. The calculation results in *ECM_MS* were compared with the results presented in [2]. Fig. 1 shows the dimensions of the test area. The main differences from [1] are as follows:

- the coil is rectangular, without rounding (a limitation of the voxel editor),
- the gap between the central strip and the yoke, circled in the figure (an *ECM_MS* limitation).
- there are no gaps at the bottom between the center strip and the yoke, and between the coil and the yoke.

Considering the research nature of the calculations, the above differences are not fundamental..

To evaluate the saturation effect, as in [2], the coil ampere-turn values of 1000, 3000, 4500 and 5000 (direct current) were chosen. The calculated quantities are the following: the average values of the **z**-component **B_z** of the magnetic flux density along the α - β line in the central strip and the γ - δ line in the yoke.

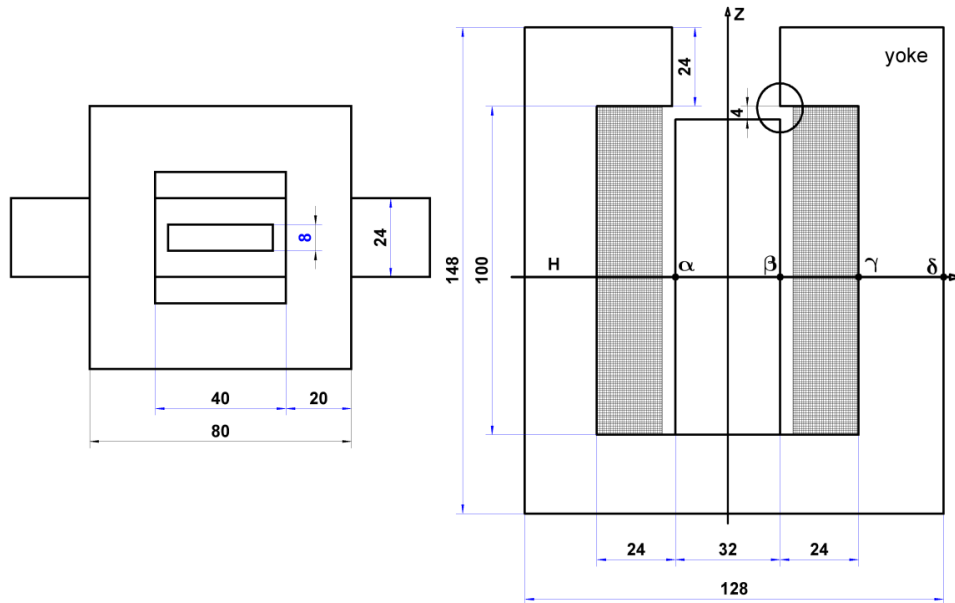


Fig 1. Shows the domain sizes for the test 1 example.

The control lines is **H**, on which the segments α - β and γ - δ are located
The gap between the central strip and the yoke is circled in the figure.

Below in fig. 2 shows a screenshot of the structure and its cross-section, prepared in a graphical editor.

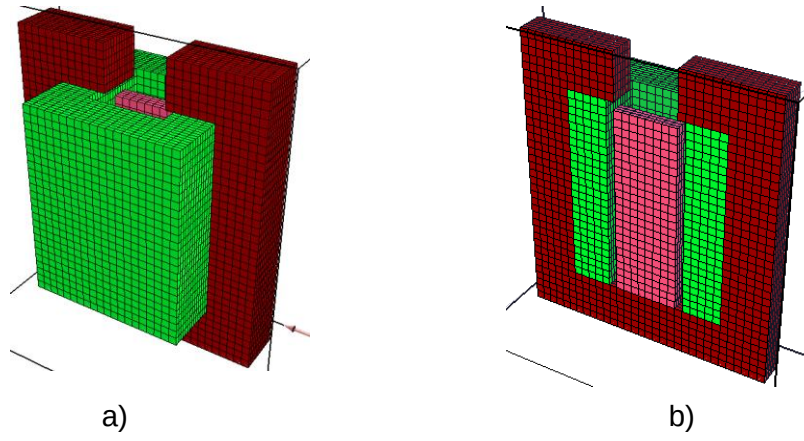


Fig 2. a) Image of the structure
b) cross-section of the structure

The values of $B(T)$ and $H(A/m)$ are given in Table 1. Fig. 3 shows the B-H curve for steel [1], [2].

Table 1. B-H curve data

	B, T	H, A/m		B, T	H, A/m
1	0.0001	0	21	1.45	1720
2	0.01	27	22	1.5	2130
3	0.025	58	23	1.55	2670
4	0.05	100	24	1.6	3480
5	0.10	153	25	1.65	4500
6	0.15	185	26	1.7	5950
7	0.2	205	27	1.75	7650
8	0.3	233	28	1.8	10100
9	0.4	255	29	1.85	13000
10	0.5	285	30	1.9	15900
11	0.6	320	31	1.95	21100
12	0.7	355	32	2.0	26300
13	0.8	405	33	2.05	32900
14	0.9	470	34	2.1	42700
15	1.0	555	35	2.15	61700
16	1.1	673	36	2.2	84300
17	1.2	836	37	2.25	110000
18	1.3	1065	38	2.3	135000
19	1.35	1220	39	2.31	140000
20	1.4	1420	40	2.33	150000

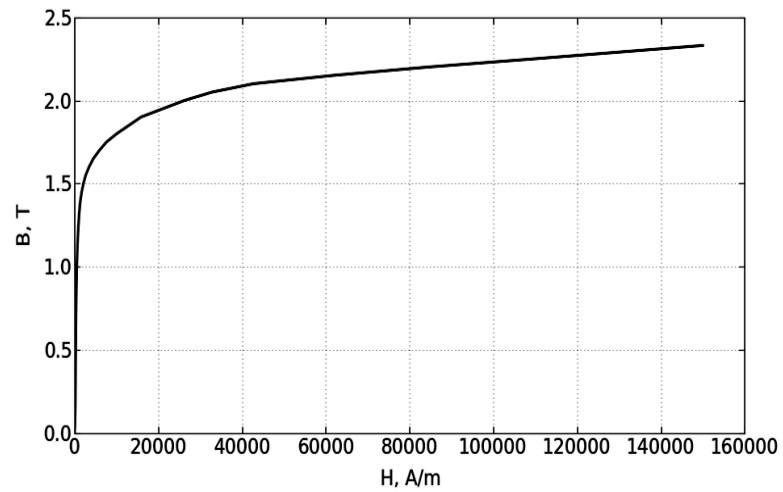


Fig 3. B-H curve graph

Below in fig. 4 shows the distribution of magnetic flux density in the cross section and a graph of the distribution of the z -component of magnetic flux density along the H line, calculated in *ECM_MS* and displayed in the **Paraview** program.

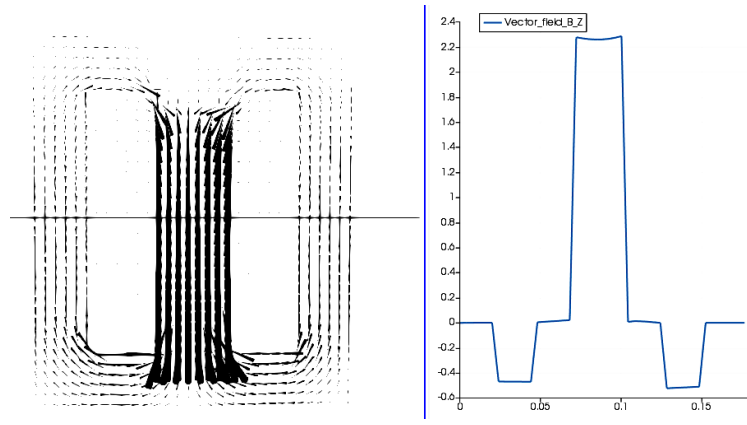


Fig. 4. Distribution of magnetic flux density in the cross section and graph of the distribution of the Z-component of magnetic flux density along the H-line

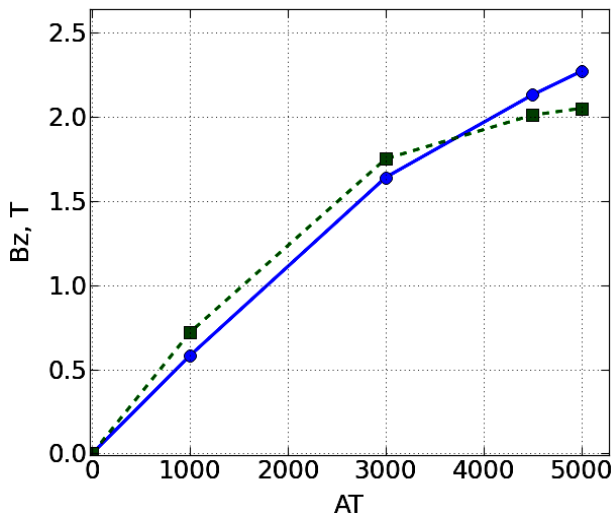
Table 2 shows the values of the z-components B_z of the average flux densities in the central strip ($\alpha-\beta$) and yoke ($\gamma-\delta$) based on the results of calculations in *ECM_MS* and given in [2]

Table 2 Z-components B_z of average flux densities in center strip ($\alpha-\beta$) and yoke ($\gamma-\delta$)

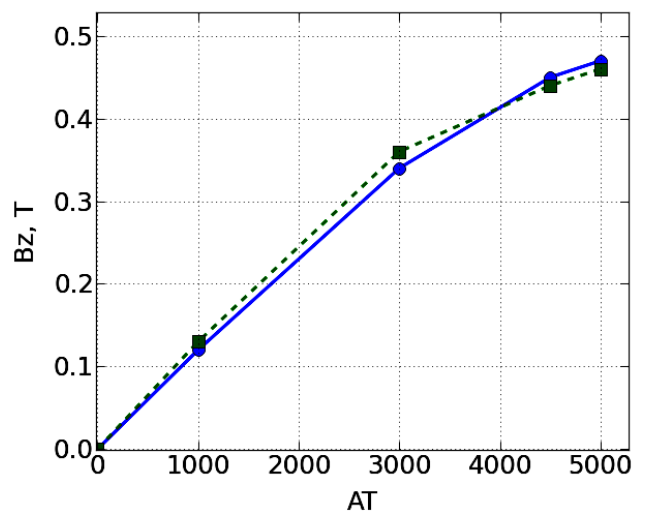
position		1000AT	3000AT	4500AT	5000AT
$\alpha-\beta$	ECM_MS	0.58	1.64	2.13	2.27
	Tabl 5 [2]*	0.72	1.75	2.01	2.05
$\gamma-\delta$	ECM_MS	0.12	0.34	0.45	0.47
	Tabl 5 [2]*	0.13	0.36	0.43	0.46

* Note: There are minor differences in the design presented in [2].

The fig.5 shows the z-component B_z of the average flux density in the central strip ($\alpha-\beta$) and the yoke ($\gamma-\delta$).



a) Z-components B_z of average flux densities in central strip (segment $\alpha-\beta$)



b) Z-components B_z of average flux densities in yoke (segment $\gamma-\delta$)

Fig. 5 The z-component B_z of average flux density in the central strip ($\alpha-\beta$) and yoke ($\gamma-\delta$).

Solid lines of the graphs correspond to calculations in the *ECM_MS*, dotted lines correspond to tabl 5 [2]

Test 2.

In the second test, the design consists of a coil with a constant current of 1800 AT and a ferromagnetic core, the $B(H)$ curve of which is shown in Fig. 3.

Fig. 6 shows information about test 2: domain sizes, an image of the structure prepared for calculations in a graphical editor, the magnetic flux density vector field calculated in *ECM_MS* displayed in **Paraview**

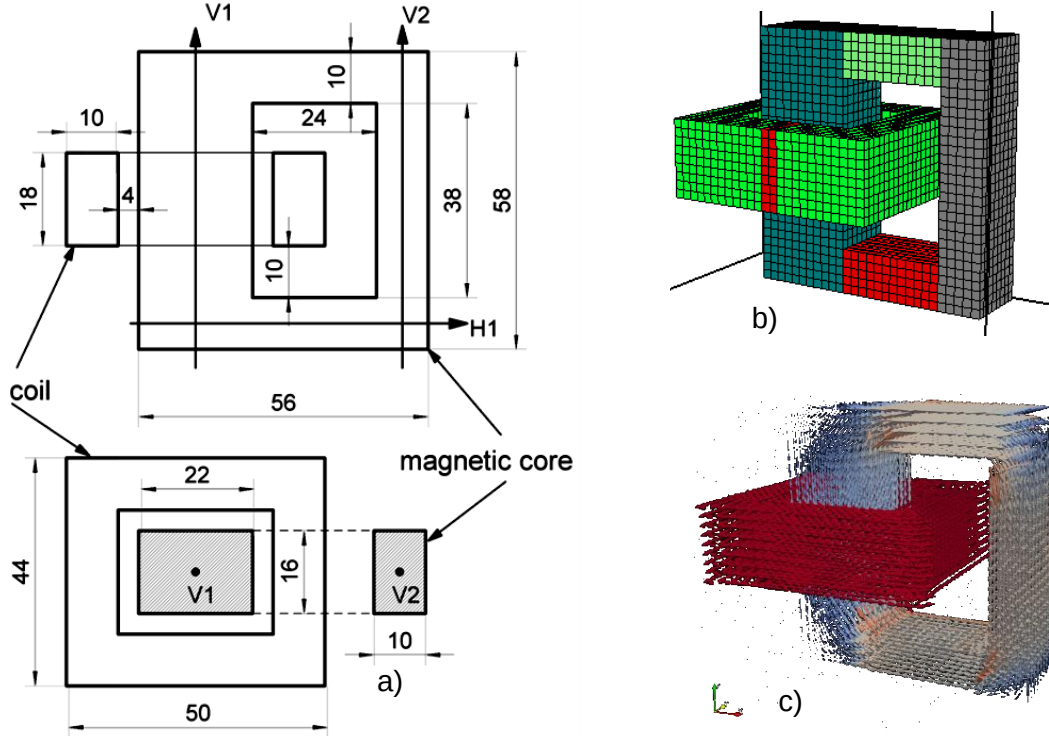


Fig.6. Test 2

- a) the domain sizes, control lines: **V1, V2, H1**;
- b) 3d image of the structure,
the core consists of four sections with identical material properties.
- c) coil current density field and magnetic flux density vector field calculated in *ECM_MS*.

The ferromagnetic core for the calculations consists of four identical parts. In Fig. 6, they are highlighted in different colors. This is due to a simplified calculation algorithm, in which nonlinear properties are averaged over the volume of the ferromagnetic. This is similar to common practice in engineering calculations (Fig. 7): sections of the ferromagnetic are replaced by equivalent magnetic resistance, and the magnetic flux is assumed to be concentrated within the core (the circuit is divided into sections with constant cross-section and material).

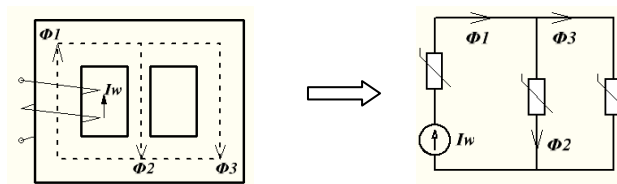


Fig. 7. In approximate calculations, the ferromagnet is divided into sections

To evaluate the results of the algorithm, a comparison was made with the results of calculating a structure with the same dimensions in the FEM software.

Below in Fig.8 are graphs of the **x**- **z**- components of the magnetic flux density along the control lines. The positive directions along the control lines are shown in Fig.6a

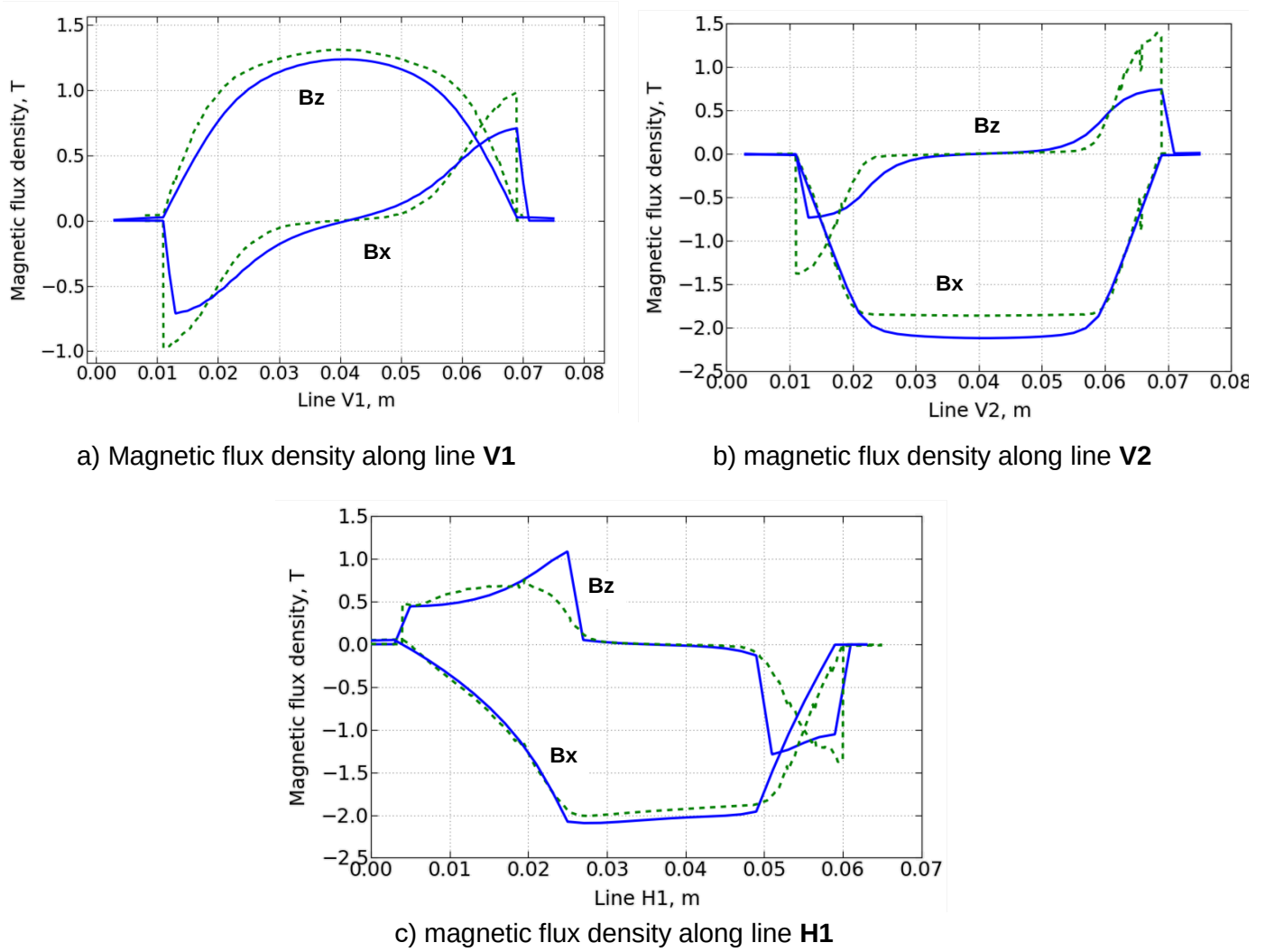


Fig.8. Graphs of the magnetic flux density. Solid lines of the graphs correspond to calculations in the *ECM_MS*, dotted lines correspond to calculations in the FEM software

Results review

In this method of equivalent electrical circuits, a contour version of the equivalent circuit is considered. Each cell face corresponds to a contour. The total number of contours in this case is greater than the number of independent contours. The system of equations formed from the contour matrix of resistances turns out to be overdetermined (has dependent rows). The Successive Over Relaxation (SOR) method was used to obtain the solution. Well-known libraries for solve overdetermined systems were also tried: *lsqr/lsmr*, *mkl_pardiso*, *lis*, but they showed worse results.

In the first test, the size of the calculation area $N_x \times N_y \times N_z = 44 \times 48 \times 48 = 101376$ cells. Cell size: $\Delta x = 4\text{mm}$, $\Delta y = 2\text{mm}$, $\Delta z = 4\text{mm}$. Total number of contours 310656. Total number of branches 317324. Number of initial SOR iterations: 2010. The number of iterations for the nonlinear part

depends on the convergence criterion and the current in the coil. With a convergence criterion of $5e-4$ (further reduction did not improve accuracy), the number of iterations at a current of 1000AT is 25, and the calculation time is 70s (including input/output time, equation generation, etc.); at a current of 5000AT, the number of iterations is 11, the calculation time is 45s.

Compared to published data (Fig. 5), this method overstates the magnetic flux density at high coil currents (it does not sufficiently account for saturation), which is explained by an overly simplified algorithm for calculating magnetic permeability.

In the second test, the size of the calculation area $N_x \times N_y \times N_z = 53 \times 38 \times 39 = 78546$ cells. Cell size: $\Delta x = \Delta y = \Delta z = 2\text{mm}$. Total number of contours: 310656. Total number of branches: 246894. Number of initial SOR iterations: 2000. Number of iterations for the nonlinear part: 11, with a convergence criterion of $1e-3$. The total computation time is on average 36s. The results were compared with calculations in FEM software. Compared to them, this method gives underestimated values when approaching the ferromagnetic boundary (Fig. 8). To improve accuracy, the ferromagnet was divided into 4 parts, and a under-relaxation coefficient of 0.7 was introduced to improve the convergence of the calculation of the nonlinear part. Without under-relaxation, convergence was lost (looping). A larger number of parts did not lead to improvement, while a smaller number reduced accuracy.

Thus, the conducted tests of the method yielded generally positive results. It can be developed in the direction of more accurate determination of the magnetic permeability of branches in the ferromagnetic region.

Bibliography

- [1]. Team Problem 20. International Compumag Society. URL:
<https://compumag.org/jsite/images/stories/TEAM/problem20.pdf>
- [2]. N. Takahashi, T. Nakata and H. Morishige" Summary of Results for Problem 20 (3-D Static Force Problem)". *COMPEL*, 1995. Vol. 14 No. 2-3 pp. 57–75,
doi: <https://doi.org/10.1108/eb010138>